

LunaSeis-1: When Balanced Apollo Seismic Detection Fails Under Cross-Station Continuous Scanning

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Abstract

Machine-learning models for lunar seismic detection are commonly developed from isolated event and background windows, but operational use requires low false-alarm rates across long continuous recordings and transfer to unseen stations. LunaSeis-1 constructs a checksum-verifiable Apollo Passive Seismic Experiment pipeline, reconciles 1,314 physical-event candidates, preserves Apollo timing/gap provenance, and evaluates lightweight detectors under event-, family-, chronology-, and station-aware controls. A 3,057-parameter tiny CNN failed a prospectively frozen continuous evaluation: it recovered 1/6 events with 2,932 false triggers over 2,591.47 station-hours. An artifact-robust successor recovered 0/3 with 847 false triggers over a second 2,651.40-hour frame. A later 2,761-parameter depthwise CNN achieved 0.9115 mean development-event recall at 0.2106 merged triggers/hour. We then froze a lower-confidence challenge containing 63 previously unexposed Grade-C natural impacts across 1,505.35 station-hours. The depthwise model recovered 12/63 at ± 180 seconds (95% exact binomial CI 0.102-0.309), produced 1,306 false triggers (0.868/hour; 95% exact Poisson CI 0.821-0.916), and retained 75.4% of the stream. These results do not establish an operational lunar detector. They show that development evaluation can substantially overstate Apollo cross-station utility and provide a reproducible benchmark, failure analysis, and open research prototype.

1. Introduction

Apollo seismic recordings remain a uniquely important record of lunar interior activity. Compact automatic detectors could assist archive analysis and, in simulation, prioritize waveform retention. However, lunar signals are emergent, instrument characteristics differ across stations and years, catalog timing semantics are imperfect, and background recordings contain substantial acquisition artifacts.

LunaSeis-1 asks whether a lightweight raw-waveform model can generalize to an excluded Apollo station while maintaining useful continuous false-alarm behavior. The primary research question is whether a compact detector selected on leakage-controlled development data retains useful event recall without persistent stream retention or excessive false triggers on prospectively frozen continuous recordings from unseen station folds. The null expectation is that development performance will not transfer reliably because station, acquisition, and catalog-timing shifts remain. No minimum operational target is claimed post hoc; recall, false triggers per station-hour, precision, and retained duration are reported jointly so that no favorable metric can conceal failure on another dimension.

This is not the first application of automated learning to Apollo seismic detection. Knapmeyer-Endrun and Hammer (2015) used hidden Markov models for Apollo 16 event detection and classification. Civilini et al. (2021) applied CNNs trained on a terrestrial station to Apollo PSE and LSPE data and motivated telemetry prioritization. Al-Qadasi and Bin Waheed (2026) evaluated compact Fourier Neural Operators on raw waveforms and spectrograms. Onodera (2024) demonstrated that short-period archive analysis materially

expands the known shallow-moonquake inventory. LunaSeis-1 therefore does not claim architectural priority. Its focus is the combined cross-station, family-aware, continuous operational stress test and transparent preservation of its negative outcome.

2. Data and provenance

Waveforms come from NASA PDS bundle `urn:nasa:pds:apollo_pse::1.0`. Labels originate from `urn:nasa:pds:apollo_seismic_event_catalog::1.0` and the corrected Onodera shallow-moonquake tables. The unified candidate registry contains 609 deep moonquakes, 623 natural impacts, 74 shallow moonquakes, and eight artificial impacts.

All downloaded products are checked against official sizes and NASA MD5 hashes. Raw products are not redistributed. MiniSEED nominal timestamps and ATT-derived timing remain separate. Gap sentinels are never silently interpolated. The candidate count is an inventory, not the number of independent or usable training examples. Waveform integrity auditing retained 1,220 candidates with at least one audited observation, marked three as questionable and ten as rejected-only, and found 81 without an archive-backed positive request.

Catalog-negative windows exclude known catalog times but can still contain uncatalogued seismic activity or acquisition artifacts; they are not equated with verified physical noise. A fixed-seed background frame sampled 928 station-days from the archive directory structure before inspecting event dates or channel completeness; 710 days passed the integrity gate. This exposed a prior sampling shortcut: S12 handcrafted-logistic performance fell from F1 0.870 and 0.284 false positives/hour on positive-conditioned days to F1 0.795 and 2.123 false positives/hour on independently selected days. Metrics from the earlier background frame were retired from model decisions.

3. Leakage controls and evaluation sequence

Four leave-one-station-out folds hold out S12, S14, S15, or S16. Physical events and known repeating families are indivisible within folds. Multiple station observations of one physical event share an event identifier, and repeating deep-moonquake families share an evaluation group. Station-held-out models cannot use normalization statistics from the test station. Architecture and threshold decisions use training/validation data only.

Three continuous frames were selected and committed before their corresponding inference runs; each is permanently consumed. The third frame was constructed only after the depthwise model and station thresholds were fixed. It contains 64 station-day observations representing 63 Grade-C natural impacts absent from prior model splits and station-day manifests. Sixteen observations were selected per station by fixed SHA-256 rank from catalog records with positive MH visibility. All 3,426 previously exposed station-days were excluded, missing products could not be replaced, and waveform integrity was audited before inference. Grade C is weaker label evidence than Grades A/B, so this frame is reported separately as a lower-confidence challenge.

4. Models and training

Comparisons include energy, STA/LTA, handcrafted logistic regression, a 3,057-parameter tiny CNN, robust variants, a 2,761-parameter depthwise CNN, and a 6,109-parameter compact dilated TCN. Neural inputs contain robust-normalized waveform values and a separate validity mask. Each 600-second window is padded or cropped to 4,096 values. Valid values are centered by the window median and scaled by the maximum of 1, 1.4826 times the median absolute deviation, and the 90th-percentile absolute deviation divided by 2.5; normalized values are clipped to ± 20 . The validity mask prevents gap sentinels from being interpreted as physical amplitude.

The selected depthwise network uses a 16-channel stem and three depthwise-separable blocks with dilations 1, 2, and 4, followed by mean and maximum temporal pooling. It was trained for 14 epochs with AdamW (learning rate 0.0007, weight decay 0.0003, fixed seed 20260714). One checkpoint and one validation-selected threshold exist per held-out station. Threshold selection required at least 90% positive-development event recall and then minimized merged development triggers; test data did not set thresholds.

5. Continuous-scanning and statistical protocol

Models scan 600-second windows at a 60-second stride. Candidate reference time is window start plus 120 seconds. Positive windows merge when adjacent inferred references are separated by at most 300 seconds. Primary one-to-one catalog matching uses ± 180 seconds, with ± 60 and ± 300 -second sensitivity analyses. An integrity-qualified window has no more than 20% missing samples. False alarms are normalized by the union of locally integrity-qualified duration. Retained duration is the union of positive 600-second windows, not the number of positive windows multiplied by window length.

A trigger matching another protected catalog reference is not counted as a false trigger, but it does not become an eligible true positive unless it belongs to the frozen target set. Recall confidence intervals use the two-sided 95% Clopper-Pearson method. False-trigger-rate intervals use a two-sided exact Poisson interval with qualified station-hours as exposure. The Poisson interval describes counting uncertainty conditional on this frame; it does not model temporal dependence among triggers. Because the three frames differ in label grade, model generation, and dates, their estimates are descriptive stress tests rather than exchangeable replicates or a pooled effectiveness estimate.

6. Results

6.1 First frozen test

Over 2,591.47 station-hours, the tiny CNN, logistic regression, and STA/LTA recovered 1/6, 1/6, and 0/6 eligible events. False-trigger rates were 1.131, 0.121, and 0.282 per hour; retained-duration fractions were 75.50%, 40.28%, and 97.92%. For the tiny CNN, recall was 0.167 (95% exact binomial CI 0.004-0.641) and the false-trigger rate was 1.131/hour (95% exact Poisson CI 1.091-1.173). The broad recall interval reflects the six-event denominator and prevents interpreting the point estimate as stable performance.

6.2 Artifact diagnosis and second frozen test

Development data revealed strong station differences in quantized plateaus, steps, and extreme-value occupancy. Robust normalization reduced the tiny CNN's second-test false-trigger rate from 0.923 to 0.319/hour and retention from 62.48% to 45.58%. Nevertheless, robust CNN, original CNN, logistic, and STA/LTA all recovered 0/3 eligible events. Logistic regression remained best by false-trigger rate at 0.185/hour. For the robust CNN, the exact 95% recall interval was 0.000-0.708 and the false-trigger-rate interval was 0.298-0.342/hour. Zero of three cannot establish zero underlying sensitivity; it establishes that the frozen frame did not demonstrate useful recall.

6.3 Later architecture development

At identical mean positive-development recall of 0.9115, the depthwise CNN produced 0.2106 merged triggers/hour, compared with 0.4102 for robust tiny CNN and 0.4558 for compact TCN. This selection occurred after both Grade-A/B tests and cannot be promoted to untouched-test evidence. A full-day CLI smoke scan also revealed persistent activation.

6.4 Event-rich Grade-C challenge

All 256 planned products (97,596,992 bytes) passed exact-size and NASA-MD5 verification. All 64 selected event-station windows passed the frozen integrity gate, yielding 1,505.35 scannable station-hours. The observations represented 63 unique physical impacts because one event was independently selected at two stations. An additional 102 catalog references on the selected days were protected from false-trigger counting.

At the primary ± 180 -second tolerance, the frozen depthwise CNN recovered 12/63 physical events (recall 0.190), generated 1,306 false triggers (0.868/hour or 20.82/day), and retained 75.38% of scannable duration. Precision excluding protected catalog triggers was 0.0091. Recall was 4/63 at ± 60 seconds and 16/63 at ± 300 seconds. The frame is consumed and no threshold or model retuning is permitted.

Frozen frame	Model	Recall (95% exact CI)	False triggers/hour (95% exact CI)	Retained
v0.1 Grade A/B	Tiny CNN	0.167 (0.004-0.641)	1.131 (1.091-1.173)	75.50%
v0.2 Grade A/B	Robust CNN	0.000 (0.000-0.708)	0.319 (0.298-0.342)	45.58%
v0.3 Grade C	Depthwise CNN	0.190 (0.102-0.309)	0.868 (0.821-0.916)	75.38%

The table does not rank the models: each row belongs to a different prospectively frozen frame, and the Grade-C row uses weaker catalog evidence. It summarizes the sequence of falsification tests.

7. Discussion

Balanced event/background windows did not predict continuous behavior. Station and acquisition regimes create powerful shortcuts, and trigger merging can hide nearly continuous positive-window activation behind apparently moderate trigger counts. Retained-duration reporting is therefore essential alongside trigger frequency.

The first two event denominators are small, but the third challenge provides a larger, deliberately weaker-label confirmation. Its 19.0% recall and 0.868 false triggers/hour reproduce the operational failure pattern despite the depthwise model's 91.15% development recall. The missed-event audit does not support a universal timing correction: score peaks occur widely over ± 2 hours and expanding tolerance mainly relabels background triggers.

The result is useful precisely because it resists a favorable demonstration narrative. Development model selection reduced merged triggers at matched development recall, yet the reserved Grade-C frame exposed both poor transfer and extensive retention. This suggests that improving architecture alone is unlikely to resolve the dominant error sources. More defensible progress requires arrival-aligned expert labels, explicit acquisition-state modeling, and evaluation dates reserved before subsequent model design.

8. Limitations

- The event-rich challenge uses Grade-C catalog labels, which are lower-confidence than the Grade-A/B development pool.
- Catalog references may not correspond to consistent physical arrivals, and many eligible windows have weak raw signal evidence.
- Artifact proxy categories are not authoritative physical diagnoses.
- Calibration is not meaningful evidence when final-test recall is zero.
- The three continuous frames are heterogeneous and cannot be pooled as independent trials of one fixed model.
- False-trigger Poisson intervals condition on observed exposure and do not account for clustered activation.
- No expert-reviewed artifact taxonomy or untouched high-confidence event-rich frame remains.
- The release model remains an experimental prototype.

9. Conclusion

LunaSeis-1 does not demonstrate an operational moonquake detector. Across three heterogeneous frozen frames totaling 6,748.22 station-hours, the model generation assigned to each frame failed the intended joint recall, false-trigger, and retention objective. The work demonstrates why apparently successful compact models can fail on continuous Apollo data and supplies a reproducible platform for improving them. Future confirmation requires better arrival-aligned labels, an explicit artifact taxonomy, and a new prospectively reserved high-confidence evaluation.

Data and code availability

Code, manifests, small checkpoints, configurations, predictions, and derived figures are prepared for public release. Raw NASA files are reconstructed from official PDS sources and are not mirrored. Release URLs and archival DOI will be inserted only after explicit publication authorization.

Reproducibility, ethics, and competing interests

All reported headline values are regenerated from frozen result JSON files, and a repository audit checks the manuscript tables against those files while calculating the stated uncertainty intervals. Configurations record seeds, fold identities, preprocessing, thresholds, matching tolerances, and consumed-frame status. The work analyzes historical robotic-mission recordings and involves no human or animal participants. The author declares no competing interests. NASA and cited catalog contributors retain attribution for source data; authorship of this paper does not imply ownership of those archives.

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Figures



Every test frame was selected and frozen before inference; consumed frames were never reused for tuning.

Figure 1. LunaSeis-1 study design. Product reconstruction, provenance-aware window construction, leakage-controlled development, and frozen continuous evaluation are separated explicitly. Consumed frames are never reused for tuning.

Frozen continuous evaluations remained below operational requirements

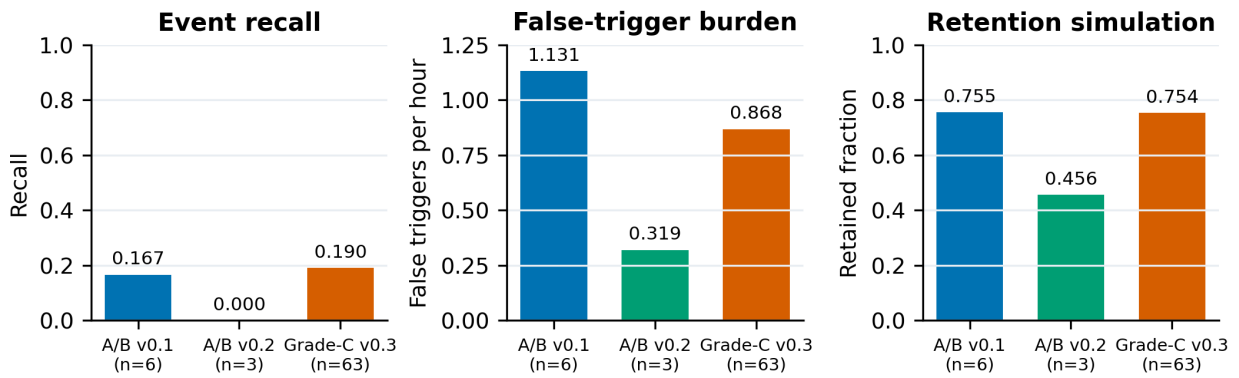


Figure 2. Primary ± 180 -second results for the three frozen continuous evaluations. Grade-A/B tests used the contemporary tiny/robust CNN; the lower-confidence Grade-C challenge used the later frozen depthwise CNN. Retained fraction measures the union of positive windows and exposes persistent activation not visible from merged-trigger counts alone.

Development selection substantially overstated operational transfer

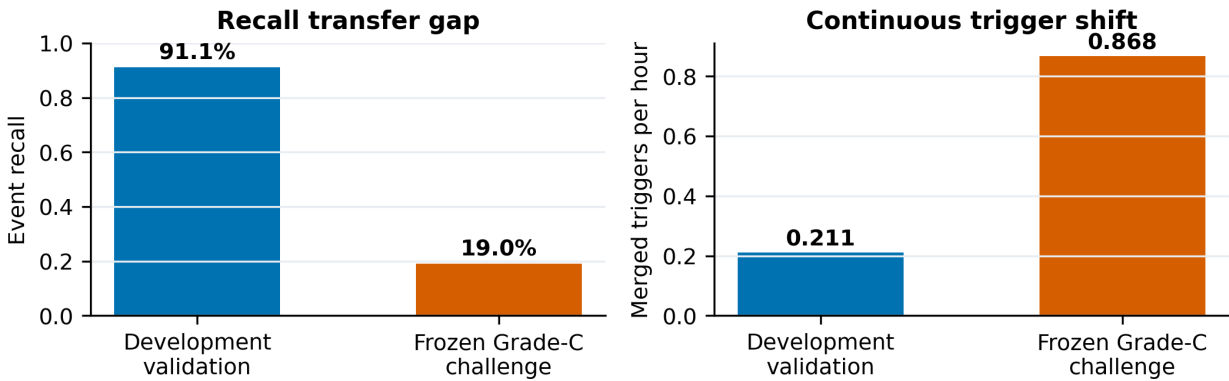


Figure 3. Transfer gap for the selected depthwise CNN. Development selection reported 91.15% mean event recall and 0.211 merged triggers/hour; the subsequent frozen Grade-C challenge yielded 19.05% recall and 0.868 false triggers/hour.

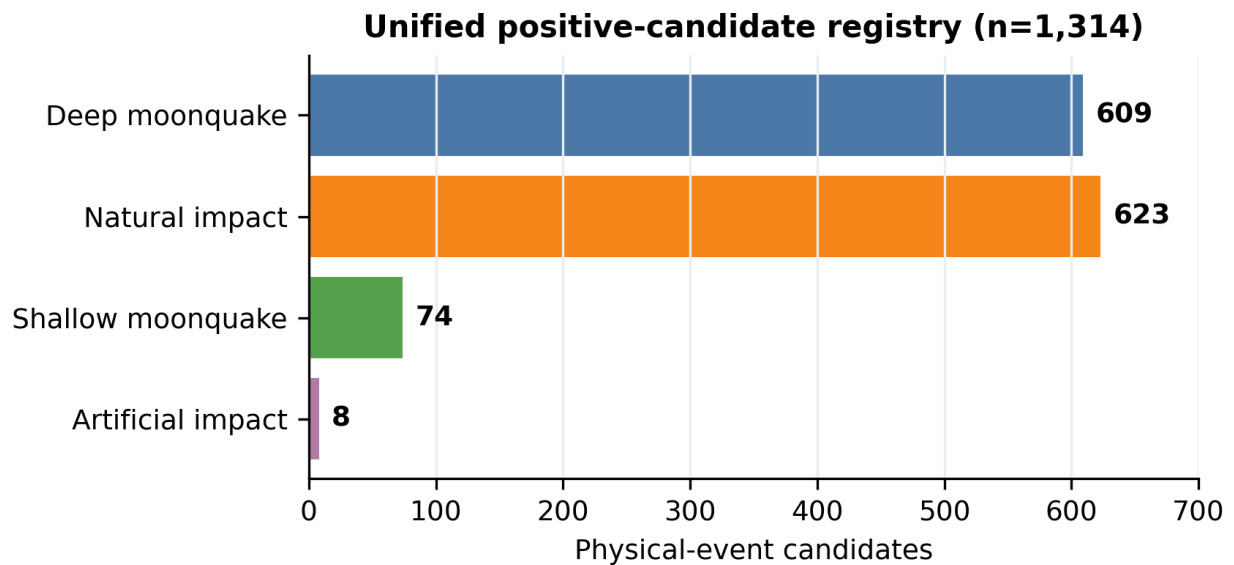


Figure 4. Physical-event composition of the unified 1,314-candidate registry. Counts are candidates with reconciled provenance, not automatically interchangeable training examples.